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TASK 1: Test Plan

1.1. Testing objective

The Umoja Learning Management System (ULMS) is a desktop application built to help manage student registrations, login authentication, course enrolment, marks capture and report generation.

Testing Goals I set out to achieve the following objectives:

- Verify that all the main features work correctly as expected.
- Identify and properly document any defects in the current system.
- Ensure data accuracy and integrity, especially in areas like marks calculation and login.
- Evaluate non-functional aspects such as performance and usability.
- Confirm that the system is stable and ready for deployment after making the necessary corrections.

Scope of Testing

In Scope:

Login, Dashboard navigation, Student Registration, Course Enrolment, Marks Capture, Average Calculation, and Report Generation.

Out of Scope:

Database persistence since the current version uses in-memory storage only, cloud integration and mobile responsiveness.

1.2 Testing Types Identification

Testing Type	Example	Approach
Functional Testing	Login Validation	Verify that the input produces the expected output
Input Validation	Marks Entry	Test valid and invalid data types, ranges, and formats
Integration Testing	Course Allocation	Test navigation and data flow between different forms
Performance Testing	Report Loading	Measure loading times and overall responsiveness
Usability Testing	Navigation	Evaluate how intuitive and user-friendly the interface is
Security Testing	Authentication	Test for unauthorized access and basic injection attempts

1.3 Testing Environment Setup

Hardware: Intel i5, CPU @ 2.80GHz, 8GB+ RAM

Software: Windows 11 Pro, Visual Studio 2026, .NET 8.0

Tools Used:

Visual Studio Debugger

OBS Studio (for evidence recording)

GitHub (version control)

Microsoft Word (test documentation)

Task 2: Test case design

Test Case ID	Description	Input	Expected Output
TC001	Successful login with valid credentials	Username = admin, Password = 1234	Login Successful and Dashboard opens
TC002	Login with invalid username	Username = Kamogelo, Password = 1234	Error message: Invalid Credentials
TC003	Login with Invalid password	Username = admin, Password = 4321	Error Message: Invalid Credentials
TC004	Login with empty fields	Username = , Password =	Error message: Fields cannot be empty.
TC005	Student Registration valid details	student ID =20240731 + Full Name=Kamogelo Phaladi + age = 21 + email= 20240731@gmail.com + select programme = Software Engineering	Student added successfully and summary display
TC006	Register student with empty fields	student ID = + Full Name= + age = + email= + select programme =	Error message: Fields should not be empty
TC007	Course Enroll student in a new course(valid)	Student ID=202470731+ Student name= Kamogelo + Selected course =Software testing + Select semester= semester 1	Success message + Course added to student
TC008	Enroll student with empty fields	Student ID= + Student name= + Selected course = + Select semester=	Error message: Fields should not be empty
TC09	Marks capture, Valid numeric input	Student ID=20240731 + Student name= Kamogelo Subject1=88, Subject2=82, Subject3=90	Average calculated correctly (86.67)
TC010	Attempt to capture empty fields	Student ID= + Student name= + Subject1= , Subject2= , Subject3=	Error message: Fields should not be empty
TC011	Attempt to capture non numeric input	Enter Student details + Subject1=pass, Subject2=fail, Subject3=pass	Validation error / Error message
TC012	Testing Maximum Average calculation	Enter Student details + Subject1=100, Subject2=100, Subject3=100	Average = 100.00
TC013	Testing Minimum Average calculation	Enter Student details + Subject1=0, Subject2=0, Subject3=0	Average = 0.00
TC014	Attempt to generate report with empty fields	Report type = + Student ID filter=	Error message: Fields should not be empty
TC015	Performance testing report display	Generate any report	Report loads within acceptable time (under 2 seconds)
TC016	Logging out	Click logout on dashboard	Directed back to Login

Declaration of Authentication



DECLARATION OF AUTHENTICITY

I Kamogelo Phaladi (FULL NAME)

hereby declare that the contents of this assessment SDT621 FA2

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

Signature: 

Date: 08/05/2026